DSA 8020 R Lab 9: Time Series Analysis

your name here, (the names of any collaborators)

Contents

1. Load the data into R. How many observations are in this time series?

Code:

```
load("RLab9.RData")
```

Answer:

2. Plot the time series and describe its features.

Code:

Answer:

3. If there is a trend component, estimate it. Explain the structure of the trend model and write down the estimated trend.

Code:

Answer:

4. Remove the estimated trend component, then plot the ACF and PACF. Based on these plots, suggest a few possible ARMA model orders.

Code:

Answer:

5. Fit the possible ARMA models identified from the previous question. Select the best model based on AIC.

Code:

Answer:

6. Diagnose if the selected time series model sufficiently accounts for temporal dependence.

Code:

Answer:

7. Use the *Arima* function in the *forecast* package to fit the first 190 observations of the original time series and perform a forecast for the last 10 observations. Plot the forecast and forecast uncertainty and comment on the forecast.

Code:

Answer: